Qualification Pack

AI Development Associate,

QP Code: NIE/ITS/Q1003

Version: 1.0

NSQF Level: 4

National Institute of Electronics and Information Technology (NIELIT) || NIELIT Bhawan, Plot No. 3, PSP Pocket, Sector-8, Dwarka
New Delhi-110077 || email:puneetnema@nielit.gov.in

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NIE/ITS/Q1003: AI Development Associate,

Brief Job Description

This programme, developed in collaboration with Intel Technologies India Pvt. Ltd., aims to empower the future workforce with essential AI skills for the digital economy. It focuses on building technical confidence, promoting responsible technology use, and enhancing employability by developing both AI competencies and relevant social skills for career growth.

Personal Attributes

An AI Development Associate should be curious, innovative, and eager to explore new technologies. They must demonstrate strong analytical and problem-solving abilities. Adaptability and a willingness to learn emerging AI tools are essential, along with a sense of ethical responsibility in applying AI solutions responsibly.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. NIE/ITS/N1035: Implementation of Basic AI Solution using Python programming language and SMART Framework,
2. NIE/ITS/N1036: Apply Machine Learning algorithms and techniques to solve industry use cases along with building an entrepreneurial mindset,
3. NIE/ITS/N1037: Realization of Projects in AI domains
4. NIE/ITS/N1038: Apply advanced AI techniques to build scalable AI solutions,
5. NIE/ITS/N1039: Solving Real-time industrial problem statements using AI,
6. NIE/ITS/N1040: Implementation of AI project in virtual environment/ OJT,
7. DGT/VSQ/N0102: Employability Skills (60 Hours)

Qualification Pack (QP) Parameters

Sector	Information Technology Sector
Sub-Sector	ARTIFICIAL INTELLIGENCE
Occupation	AI DEVELOPMENT

Qualification Pack

Country	India
NSQF Level	4
Credits	19
Aligned to NCO/ISCO/ISIC Code	4
Minimum Educational Qualification & Experience	12th grade Pass (or equivalent) OR Completed 2nd year of the 3-year diploma after 10 (in CS/IT/EC/EE/ allied branches) with NA of experience OR 12th grade Pass (who have undergone ARTIFICIAL INTELLIGENCE course as Vocational subjects in class 12th, Such students shall be exempted from the first 2 NOSs. i. Implementation of Basic AI Solution using Python programming language and SMART Framework. ii. Apply Machine Learning algorithms and techniques to solve industry use cases along with building an entrepreneurial mindset) OR Previous relevant Qualification of NSQF Level (3) with 3 Years of experience experience in relevant field
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	06/02/2029
NSQC Approval Date	06/02/2026
Version	1.0
Reference code on NQR	QG-04-IT-00141-2026-V2-NIELIT
NQR Version	2

NIE/ITS/N1035: Implementation of Basic AI Solution using Python programming language and SMART Framework,

Description

This NOS describes the competencies required to understand Artificial Intelligence concepts and implement basic AI solutions using Python programming. It includes defining AI problems using the SMART framework, following the AI project cycle, performing data analysis with Python libraries and Orange Data Mining Tool, and visualizing results through dashboards. The NOS emphasizes ethical, inclusive, and application-oriented AI practices aligned with emerging technology requirements.

Scope

The scope covers the following :

- This NOS enables learners to understand AI fundamentals, apply the SMART framework, follow the AI project cycle, and implement basic AI solutions using Python. It covers Python libraries, Orange Data Mining, AI applications, ethical AI concepts, and deployment of insights through data visualization dashboards using Tableau Public.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Define AI problem statements using the SMART framework and identify suitable real-world applications.

PC2. Follow the AI project cycle to acquire, analyze, model, and evaluate data for basic AI solutions.

PC3. Use Python programming and relevant libraries (NumPy, Pandas, Scikit-learn) for data analysis and model development.

PC4. Apply Orange Data Mining Tool for exploratory data analysis and simple machine learning workflows.

PC5. Visualize, interpret, and communicate AI outcomes using dashboards and charts through Tableau Public.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Fundamental concepts of Artificial Intelligence, its applications, limitations, and ethical considerations.

KU2. The SMART framework and its role in defining clear, measurable, and achievable AI objectives.

KU3. Stages of the AI project cycle and their application in developing AI-based solutions.

KU4. Basics of Python programming and the use of Python libraries such as NumPy, Pandas, and Scikit-learn.

KU5. Data analysis, visualization, and deployment concepts using tools like Orange Data Mining Tool and Tableau Public.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Analyze real-world problems and identify opportunities for applying basic AI solutions.

GS2. Write and execute simple Python programs for data handling, analysis, and model development.

GS3. Use analytical tools to explore data, interpret results, and draw meaningful insights.

GS4. Communicate AI concepts, findings, and visualizations effectively to technical and non-technical stakeholders.

GS5. Apply ethical, inclusive, and responsible practices while working with AI technologies and data.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	25	**10**	**-**	**-**
PC1. Define AI problem statements using the SMART framework and identify suitable real-world applications.	-	-	-	-
PC2. Follow the AI project cycle to acquire, analyze, model, and evaluate data for basic AI solutions.	-	-	-	-
PC3. Use Python programming and relevant libraries (NumPy, Pandas, Scikit-learn) for data analysis and model development.	-	-	-	-
PC4. Apply Orange Data Mining Tool for exploratory data analysis and simple machine learning workflows.	-	-	-	-
PC5. Visualize, interpret, and communicate AI outcomes using dashboards and charts through Tableau Public.	-	-	-	-
NOS Total	**25**	**10**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N1035
NOS Name	Implementation of Basic AI Solution using Python programming language and SMART Framework,
Sector	Information Technology Sector
Sub-Sector	Artificial Intelligence
Occupation	AI Development
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2029
NSQC Clearance Date	06/02/2026

NIE/ITS/N1036: Apply Machine Learning algorithms and techniques to solve industry use cases along with building an entrepreneurial mindset,

Description

This NOS describes the competencies required to apply Machine Learning algorithms and AI techniques using Python to solve industry-relevant use cases. It includes understanding and using Python libraries and AI models, applying Design Thinking and Systems Thinking, identifying and mitigating AI bias, and developing an entrepreneurial mindset for opportunity identification, innovation, and future job creation.

Scope

The scope covers the following :

- This NOS covers the skills and knowledge required to apply Machine Learning algorithms using Python to solve real-world industry problems. It includes using Python libraries and AI models, applying Design Thinking and Systems Thinking, understanding and addressing AI bias, and developing an entrepreneurial mindset to identify opportunities, innovate solutions, and align AI applications with future employment and business needs.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Identify and analyze industry use cases suitable for Machine Learning solutions using Python.

PC2. Select and apply appropriate AI and Machine Learning models to solve real-world problems.

PC3. Use Python libraries to preprocess data, train models, and evaluate outcomes effectively.

PC4. Apply Design Thinking and Systems Thinking principles while recognizing and mitigating AI bias in AI projects.

PC5. Communicate technical solutions clearly, collaborate effectively, and demonstrate initiative and entrepreneurial thinking in professional environments.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Core concepts of Machine Learning, common algorithms, and their application in industry use cases.

KU2. Python programming fundamentals and key libraries used for data analysis and model development.

KU3. Principles of Design Thinking and Systems Thinking in the context of AI solution development.

KU4. Sources, impacts, and mitigation strategies related to AI bias and ethical AI practices.

KU5. The concept of an entrepreneurial mindset and its importance for opportunity identification, innovation, and future employment.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Analyze industry problems and translate them into Machine Learning problem statements.

GS2. Develop, test, and refine AI models using Python in a structured and systematic manner.

GS3. Apply critical thinking and creativity while using Design Thinking and Systems Thinking approaches.

GS4. Identify ethical concerns and biases in data and models and take corrective actions.

GS5. Communicate technical solutions clearly, collaborate effectively, and demonstrate initiative and entrepreneurial thinking in professional environments.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	50	**22**	**-**	**-**
PC1. Identify and analyze industry use cases suitable for Machine Learning solutions using Python.	-	-	-	-
PC2. Select and apply appropriate AI and Machine Learning models to solve real-world problems.	-	-	-	-
PC3. Use Python libraries to preprocess data, train models, and evaluate outcomes effectively.	-	-	-	-
PC4. Apply Design Thinking and Systems Thinking principles while recognizing and mitigating AI bias in AI projects.	-	-	-	-
PC5. Communicate technical solutions clearly, collaborate effectively, and demonstrate initiative and entrepreneurial thinking in professional environments.	-	-	-	-
NOS Total	**50**	**22**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N1036
NOS Name	Apply Machine Learning algorithms and techniques to solve industry use cases along with building an entrepreneurial mindset,
Sector	Information Technology Sector
Sub-Sector	Artificial Intelligence
Occupation	AI Development
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2029
NSQC Clearance Date	06/02/2026

NIE/ITS/N1037: Realization of Projects in AI domains

Description

This NOS defines the competencies required to plan and execute AI projects using supervised, unsupervised, and reinforcement learning. It covers key AI domains such as Computer Vision, Data Analysis, NLP, and Generative AI, along with AI ethics, social emotional skills, and identification of project pitfalls across the AI project cycle.

Scope

The scope covers the following :

- This NOS covers the ability to plan, develop, and evaluate AI projects using appropriate learning paradigms and AI technologies. It includes understanding core AI domains and their real-world applications, applying ethical principles and best practices, relating Social Emotional Skills to AI project work, and identifying common project pitfalls at different stages of the AI project cycle to improve project outcomes and decision-making.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Distinguish between supervised, unsupervised, and reinforcement learning approaches and select them appropriately for AI projects.

PC2. Apply AI techniques in domains such as Computer Vision, Statistical Data Analysis, Natural Language Processing, and Generative AI.

PC3. Integrate ethical principles and social–emotional skills while planning and executing AI projects.

PC4. Identify and address common project pitfalls at different stages of the AI project cycle.

PC5. Evaluate AI project outcomes critically to ensure responsible, effective, and practical AI solutions.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Concepts, characteristics, and differences among supervised, unsupervised, and reinforcement learning.

KU2. Fundamentals of Computer Vision, Statistical Data Analysis, Natural Language Processing, and Generative AI with their current applications.

KU3. The five pillars of Social Emotional Skills and their relationship with Emotional Intelligence.

KU4. Principles of AI ethics, responsible AI, and best practices in AI development.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Apply critical thinking to analyze AI problems and select appropriate learning techniques.

GS2. Collaborate effectively in teams while demonstrating social emotional awareness and professionalism.

GS3. Identify ethical risks and biases in AI projects and apply responsible decision-making.

GS4. Manage AI projects systematically by anticipating challenges and avoiding common project pitfalls.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	60	**23**	**-**	**-**
PC1. Distinguish between supervised, unsupervised, and reinforcement learning approaches and select them appropriately for AI projects.	-	-	-	-
PC2. Apply AI techniques in domains such as Computer Vision, Statistical Data Analysis, Natural Language Processing, and Generative AI.	-	-	-	-
PC3. Integrate ethical principles and social–emotional skills while planning and executing AI projects.	-	-	-	-
PC4. Identify and address common project pitfalls at different stages of the AI project cycle.	-	-	-	-
PC5. Evaluate AI project outcomes critically to ensure responsible, effective, and practical AI solutions.	-	-	-	-
NOS Total	**60**	**23**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N1037
NOS Name	Realization of Projects in AI domains
Sector	Information Technology Sector
Sub-Sector	Artificial Intelligence
Occupation	AI Development
NSQF Level	4
Credits	3
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2029
NSQC Clearance Date	06/02/2026

NIE/ITS/N1038: Apply advanced AI techniques to build scalable AI solutions,

Description

This NOS enables learners to apply advanced AI techniques, including generative AI, no-code AI agents, and optimized inferencing using OpenVINO and Intel oneAPI, to build scalable, high-performance AI solutions while considering ethical and societal impacts.

Scope

The scope covers the following :

- This NOS covers the application of generative AI models, prompt engineering, development of no-code AI agent workflows, and deployment of scalable AI inferencing solutions using OpenVINO and Intel oneAPI for real-world business and industry use cases.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Identify appropriate industry use cases for applying generative AI models (text, image, audio, or video).

PC2. Design, test, and refine prompts by defining clear goals, context, and constraints to improve AI-generated outputs.

PC3. Develop no-code AI agent workflows using goals, tools, memory, and automation to streamline business processes.

PC4. Implement and optimize AI inferencing pipelines using OpenVINO and Intel's oneAPI for scalable and high-performance solutions.

PC5. Evaluate, optimize, and document AI solutions based on performance metrics, reliability, and deployment outcomes.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Understanding generative AI models (text, image, audio, and video) and their applications across various industry domains.

KU2. Applying prompt engineering principles such as structure, clarity, context, and constraints to influence AI-generated outputs.

KU3. Using AI agent architectures involving goals, tools, memory, and automation to streamline and optimize business workflows.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Apply analytical and problem-solving skills to select, design, and optimize AI solutions for real-world industry scenarios.

GS2. Demonstrate professional judgment in the responsible, ethical, and secure use of AI technologies and data.

GS3. Communicate effectively and document AI workflows, results, and improvements for collaboration, deployment, and scalability.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	40	**22**	**-**	**-**
PC1. Identify appropriate industry use cases for applying generative AI models (text, image, audio, or video).	-	-	-	-
PC2. Design, test, and refine prompts by defining clear goals, context, and constraints to improve AI-generated outputs.	-	-	-	-
PC3. Develop no-code AI agent workflows using goals, tools, memory, and automation to streamline business processes.	-	-	-	-
PC4. Implement and optimize AI inferencing pipelines using OpenVINO and Intel's oneAPI for scalable and high-performance solutions.	-	-	-	-
PC5. Evaluate, optimize, and document AI solutions based on performance metrics, reliability, and deployment outcomes.	-	-	-	-
NOS Total	**40**	**22**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N1038
NOS Name	Apply advanced AI techniques to build scalable AI solutions,
Sector	Information Technology Sector
Sub-Sector	
Occupation	AI DEVELOPMENT
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2029
NSQC Clearance Date	06/02/2026

NIE/ITS/N1039: Solving Real-time industrial problem statements using AI,

Description

This NOS focuses on enabling learners to analyze problem statements to identify the appropriate analytical focus, qualify and integrate data from multiple sources, and evaluate data attributes for suitability in AI applications. It emphasizes assessing model outcomes with respect to bias and variance, and defining, selecting, and qualifying AI models to ensure accuracy, reliability, and responsible deployment in real-world scenarios.

Scope

The scope covers the following :

- This NOS covers problem analysis to determine the appropriate analytical focus, qualification and evaluation of data from multiple sources, assessment of data attributes, analysis of bias and variance in results, and definition and qualification of AI models for reliable and responsible use.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Identify and define the appropriate analytical focus while interpreting a given problem statement.

PC2. Collect, qualify, and integrate data from multiple sources for AI or analytical tasks.

PC3. Evaluate data attributes such as quality, relevance, completeness, and consistency.

PC4. Assess analysis or model results for bias and variance and apply corrective measures where required.

PC5. Define, select, and qualify suitable AI models based on data characteristics and desired outcomes.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Ability to analyze problem statements and identify the correct analytical or AI focus.

KU2. Ability to source, qualify, and integrate data from multiple sources for analysis or AI tasks.

KU3. Ability to assess data attributes such as quality, relevance, completeness, and consistency.

KU4. Ability to interpret bias and variance in results and their impact on AI model selection and performance.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Apply analytical and critical-thinking skills to evaluate problems, data sources, and AI model outcomes.

GS2. Demonstrate professional judgment in identifying bias, ensuring data quality, and promoting responsible AI practices.

GS3. Communicate analysis results and model decisions clearly through documentation and collaboration with stakeholders.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	25	**13**	**-**	**-**
PC1. Identify and define the appropriate analytical focus while interpreting a given problem statement.	-	-	-	-
PC2. Collect, qualify, and integrate data from multiple sources for AI or analytical tasks.	-	-	-	-
PC3. Evaluate data attributes such as quality, relevance, completeness, and consistency.	-	-	-	-
PC4. Assess analysis or model results for bias and variance and apply corrective measures where required.	-	-	-	-
PC5. Define, select, and qualify suitable AI models based on data characteristics and desired outcomes.	-	-	-	-
NOS Total	**25**	**13**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N1039
NOS Name	Solving Real-time industrial problem statements using AI,
Sector	Information Technology Sector
Sub-Sector	
Occupation	AI DEVELOPMENT
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2029
NSQC Clearance Date	06/02/2026

NIE/ITS/N1040: Implementation of AI project in virtual environment/ OJT,

Description

The AI Development Associate project focuses on building foundational skills in artificial intelligence, including data preparation, model development, training, and evaluation. It enables learners to apply machine learning algorithms for problem-solving across domains such as image, text, and predictive analytics. The project also emphasizes ethical AI practices and model deployment techniques.

Scope

The scope covers the following :

- The scope of the AI Development Associate project includes understanding and applying fundamental concepts of Artificial Intelligence such as machine learning, deep learning, and natural language processing. It covers designing, developing, and testing AI models for various real-world applications.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Identify and scope a real-world industrial or social problem suitable for an AI-based project.

PC2. Collect, preprocess, and analyze relevant data to support the project requirements.

PC3. Design, develop, and implement an appropriate AI/ML solution using suitable tools and techniques.

PC4. Test, validate, and optimize the solution while addressing ethical, bias, and data privacy concerns.

PC5. Document, present, and demonstrate the project, clearly highlighting outcomes, impact, and learnings.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Understanding of AI concepts, types, and applications, including machine learning, deep learning, and natural language processing.

KU2. Knowledge of data collection, cleaning, normalization, and preparation techniques essential for training AI models.

KU3. Familiarity with key AI development platforms, libraries, and tools such as Python, TensorFlow, Keras, and Scikit-learn.

KU4. Understanding of AI model building processes, including model training, testing, evaluation metrics, and optimization techniques.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Ability to analyze data-driven problems and develop effective AI-based solutions using critical thinking approaches.

GS2. Skilled in programming languages, AI tools, and machine learning techniques for designing and developing intelligent systems.

GS3. Competent in handling, processing, and analyzing large datasets essential for training AI models.

GS4. Capable of working in teams and effectively communicating technical concepts and project outcomes to diverse audiences.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	**-**	**-**	**-**
PC1. Identify and scope a real-world industrial or social problem suitable for an AI-based project.	-	-	-	-
PC2. Collect, preprocess, and analyze relevant data to support the project requirements.	-	-	-	-
PC3. Design, develop, and implement an appropriate AI/ML solution using suitable tools and techniques.	-	-	-	-
PC4. Test, validate, and optimize the solution while addressing ethical, bias, and data privacy concerns.	-	-	-	-
PC5. Document, present, and demonstrate the project, clearly highlighting outcomes, impact, and learnings.	-	-	-	-
NOS Total	**-**	**-**	**30**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N1040
NOS Name	Implementation of AI project in virtual environment/ OJT,
Sector	Information Technology Sector
Sub-Sector	
Occupation	AI DEVELOPMENT
NSQF Level	4
Credits	8
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2029
NSQC Clearance Date	06/02/2026

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. identify employability skills required for jobs in various industries

PC2. identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.

PC4. follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC5. recognize the significance of 21st Century Skills for employment

PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC7. use basic English for everyday conversation in different contexts, in person and over the telephone

PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English

PC9. write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

PC10. understand the difference between job and career

PC11. prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings

PC13. work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC14. communicate and behave appropriately with all genders and PwD

PC15. escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC16. select financial institutions, products and services as per requirement

PC17. carry out offline and online financial transactions, safely and securely

PC18. identify common components of salary and compute income, expenses, taxes, investments etc

PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC20. operate digital devices and carry out basic internet operations securely and safely

PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively

PC22. use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research

PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion

PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

PC26. identify different types of customers

PC27. identify and respond to customer requests and needs in a professional manner.

PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

GS3. behave politely and appropriately with all
GS4. how to work in a virtual mode
GS5. perform calculations efficiently
GS6. solve problems effectively
GS7. pay attention to details
GS8. manage time efficiently
GS9. maintain hygiene and sanitization to avoid infection

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Introduction to Employability Skills	**1**	**1**	**-**	**-**
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
Constitutional values – Citizenship	**1**	**1**	**-**	**-**
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
Becoming a Professional in the 21st Century	**2**	**4**	**-**	**-**
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
Basic English Skills	**2**	**3**	**-**	**-**
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
Career Development & Goal Setting	**1**	**2**	**-**	**-**

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
Communication Skills	**2**	**2**	**-**	**-**
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
Diversity & Inclusion	**1**	**2**	**-**	**-**
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
Financial and Legal Literacy	**2**	**3**	**-**	**-**
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
Essential Digital Skills	**3**	**4**	**-**	**-**
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Entrepreneurship	**2**	**3**	**-**	**-**
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
Customer Service	**1**	**2**	**-**	**-**
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
Getting ready for apprenticeship & Jobs	**2**	**3**	**-**	**-**
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	**20**	**30**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

Assessment of the qualification evaluates candidates to ascertain that they can integrate knowledge, skills and values for carrying out relevant tasks as per the defined learning outcomes and assessment criteria.

The underlying principle of assessment is fairness and transparency. The evidence of the outcomes and assessment criteria. Competence acquired by the candidate can be obtained by conducting Theory (Online), Practical assessment, internal assessment, Project/Presentation/ Assignment, Major Project. The emphasis is on the practical demonstration of skills & knowledge gained by the candidate through the training. Each OUTCOME is assessed & marked separately. A candidate is required to pass all OUTCOMES individually based on the passing criteria.

About Examination Pattern:

1. The question papers for the theory and practical exams are set by the Examination wing (assessor) of

NIELIT HQS.

2. The assessor assigns roll number.

3. The assessor carries out theory online assessments through remote proctoring methodology. Theory examination would be conducted online and the paper comprise of MCQ. Conduct of assessment are through trained proctors. Once the test begins, remote proctors have full access to candidate's video feeds and computer screens. Proctors authenticate the candidate based on registration details, pre-test image captured and I- card in possession of the candidate. Proctors can chat with candidates or give warnings to candidates. Proctors can also take screenshots, terminate a specific user's test session, or re-authenticate candidates based on video feeds.

4. An External Examiner/ Observer may be deployed including NIELIT officials for evaluation of Practical examination/ internal assessment / Project/ Presentation/. Major Project (if applicable) would be evaluated preferably by external/ subject expert including NIELIT officials.

5. Pass percentage would be 50% marks in each component.

6. Candidates may apply for re-examination within the validity of registration (only in the assessment component in which the candidate failed).

7. For re-examination prescribed examination fee is required to be paid by the candidate only for the assessment component in which the candidate wants to reappear.

8. There would be no exemption for any paper/module for candidates having similar qualifications or skills.

9. The examination will be conducted in English language only.

Quality assurance activities: A pool of questions is created by a subject matter expert and moderated by other SME. Test rules are set beforehand. Random set of questions which are according to syllabus appears which may differ from candidate to candidate. Confidentiality and impartiality are maintained during all the examination and evaluation processes.

Minimum Aggregate Passing % at QP Level : 50

(**Please note**: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
NIE/ITS/N1035.Implementation of Basic AI Solution using Python programming language and SMART Framework,	25	10	-	-	35	10
NIE/ITS/N1036.Apply Machine Learning algorithms and techniques to solve industry use cases along with building an entrepreneurial mindset,	50	22	-	-	72	20
NIE/ITS/N1037.Realization of Projects in AI domains	60	23	-	-	83	23
NIE/ITS/N1038.Apply advanced AI techniques to build scalable AI solutions,	40	22	-	-	62	18
NIE/ITS/N1039.Solving Real-time industrial problem statements using AI,	25	13	-	-	38	11
NIE/ITS/N1040.Implementation of AI project in virtual environment/ OJT,	-	-	30	-	30	9
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	9
Total	**220**	**120**	**30**	**-**	**370**	**100**

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
AA	Assessment Agency
AB	Awarding Body
NCO	National Classification of Occupations
NCrF	National Credit Framework
NOS	National Occupational Standard(s)
NQR	National Qualification Register
NSQF	National Skills Qualifications Framework

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.
Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.

Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.
National Occupational Standard	NOS define the measurable performance outcomes required from an individual engaged in a particular task. They list down what an individual performing that task should know and also do.
Qualification	A formal outcome of an assessment and validation process which is obtained when a competent body determines that an individual has achieved learning outcomes to given standards
Qualification File	A Qualification File is a template designed to capture necessary information of a Qualification from the perspective of NSQF compliance. The Qualification File will be normally submitted by the awarding body for the qualification.
Sector	A grouping of professional activities on the basis of their main economic function, product, service or technology
Long Term Training	Long-term skilling means any vocational training program undertaken for a year and above. https://ncvet.gov.in/sites/default/files/NCVET.pdf